



HOUSE PRICE PREDICTION REPORT

MACHINE LEARNING - INTERNSHIP REPORT

Introduction

House price prediction is a common application of machine learning in the real estate industry. Accurate price estimation helps buyers, sellers, and investors make informed decisions. In this project, machine learning algorithms were used to analyze house features and predict prices

Objectives

- Perform Exploratory Data Analysis (EDA)
- Handle missing values
- Apply feature engineering
- Train multiple regression models
- Compare model performance using RMSE and MAE
- Save the best model for future prediction

Dataset Description

The dataset contains various features of houses such as:

- Area
- Bedrooms
- Bathrooms
- Stories
- Parking
- Furnishing Status

The target variable is:

- price

Tools and Libraries Used

- Python
- Jupyter Notebook
- pandas

- NumPy
- Matplotlib
- Seaborn
- scikit-learn

Methodology

1. Data Loading
2. Data Cleaning
3. Feature Engineering
4. Train-Test Split
5. Model Training
6. Model Evaluation
7. Model Saving

Models Used

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor

Evaluation Metrics

- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)
- Lower values indicate better model performance.

Conclusion

This project successfully developed a machine learning model to predict house prices. Among the tested models, Gradient Boosting Regressor provided the best results. The final model was saved and can be used to predict prices for new houses.